TMAP-Bulk + OMAP-Shilov Paired Sister Principles v0.1 — Bulk-vs-Shilov Substrate-Mechanism Distinction

Keeper (Casey-directive synthesis 2026-05-27 Wednesday PM)

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1. Casey directive context

Casey 2026-05-27 PM, in response to Keeper question #1 (bulk-vs-Shilov substrate-mechanism distinction):

> *"YES, two regions for different types of particles, of course. Neutrinos and Electrons and their brethren are light, fundamental, and all closely related to the S^1 Shilov Boundary, the confined composite heavy particles are 3D constructions from a different region, so make the work and papers show the relationship."*

This elevates Lyra's afternoon surface finding (echoed by Grace cross-sector observations Wednesday) to a substantive structural framework: the substrate operates DIFFERENTLY in two regions, using different mathematical ingredient sets for each particle type class.

2. The bulk-vs-Shilov substrate-mechanism distinction

2.1 Two regions

D_IV^5 has two substrate-natural regions:

Region	Geometric structure	Particle class	Mathematical ingredient set
Shilov boundary $S^4 \times S^1$	Distinguished maximum-modulus locus; Pin(2) double cover acting on S^1	Light / fundamental: leptons (e, μ, τ, ν_e, ν_μ, ν_τ), photons, gauge bosons	Ogg supersingular primes + Moonshine structure: {17, 19, 23, 71, …}
Bulk D_IV^5 interior	Holomorphic interior of bounded symmetric domain; 3D Euclidean projection per Bergman 7/2 weighting	Heavy / composite / 3D-constructed: quarks (u, d, c, s, t, b), nucleons (p, n), nuclei	Mersenne primes + cyclotomic chain: {3, 7, 31, 127, …}

2.2 Why two regions

Per Tuesday morning Bose-Fermi ↔ S^4-S^1 mapping (13/13 fermion observable empirical): - Fermions live on Pin(2) cover of S^1 Shilov factor - This places leptons (fundamental fermions) on Shilov S^1 — confirming Casey's framing

Per Wednesday morning Lyra Track DC + Elie Toy 3554 + Grace 6-instance extension: - Cal #139 cyclotomic chain = q=2 specialization of standard q-integers - Substrate's chain {rank, N_c, n_C, g} = {2, 3, 5, 7} uses Mersenne primes M_p = 2^p − 1 - This operates in BULK substrate-mechanism territory

Per Casey directive Tuesday afternoon “S^1 enacts physics”: - Shilov $S^4 \times S^1$ enacts physics; bulk D_IV^5 is rigid stable host - DIFFERENT particle classes have DIFFERENT enaction modes

The 9-element substrate operational arithmetic set Grace surfaced (INV-5195) — {BST primaries + extended Casimirs + Ogg supersingular} — is the UNION of two regional substrate-mechanism sets, not a single unified set.

2.3 Casey’s framing applies

> *“…the confined composite heavy particles are 3D constructions from a different region…”*

Bulk region has 3D-construction character — fits with Bergman 7/2 weight asymptotic-projection from D_IV^5 holomorphic 5-complex-dim to observable 4D physical spacetime, with effective 3D Euclidean structure for bound states (hydrogen 1s spherical orbital, nucleon confinement envelope, etc.).

Shilov region has fundamental-not-composite character — leptons + photons + gauge bosons are point-like irreducible substrate-residents on the boundary.

3. TMAP-Bulk principle (Casey-named candidate, Bulk region)

Statement (v0.1): The substrate-mechanism for bulk-region (quark/composite-heavy/3D-constructed) particle observables operates via Mersenne primes and cyclotomic chain structure at q=2 specialization.

Supporting evidence (Wednesday substantive findings): - m_b/m_d = g · M_g = 7 · 127 = 889 (Lyra finding; SUBSTANTIVE Mersenne substrate-natural match for bottom-down quark skip ratio) - Cal #139 chain at q=2 organizes substrate’s Mersenne structure - K59 RATIFIED Reed-Solomon coding on GF(2^g) = GF(128) provides bulk substrate-computational backbone - Cyclotomic chain forcing (rank, N_c) → (n_C, g) via Mersenne lifts operates in bulk substrate - Cross-sector 3rd-generation X=g pattern (lepton g^2·Ogg71 + quark g·M_g) — the chain-level X=g applies to both regions but with different prime-set ingredients

Falsification gate: any quark-sector observable that operates via Ogg/Moonshine structure (not Mersenne) would refine TMAP-Bulk scope; full universal-falsification would require quark observables exclusively using Shilov-region ingredients.

Lyra v0.1 quark sector test today: partially falsified TMAP-universal but m_b/m_d substantive Mersenne match — exactly the pattern TMAP-Bulk predicts (Mersenne-in-bulk, not universal).

4. OMAP-Shilov principle (Casey-named candidate, Shilov region)

Statement (v0.1): The substrate-mechanism for Shilov-boundary (lepton/light/fundamental) particle observables operates via Ogg supersingular primes and Monster Moonshine structure on the Pin(2) cover of S^1.

Supporting evidence (Wednesday substantive findings): - $m\_\tau/m\_e = g^2 \cdot \text{Ogg71} = 49 \cdot 71 = 3479$ (Grace + Lyra finding; T2003 RATIFIED; both factors substrate-natural — g^2 from chain level X=g, Ogg71 = largest Ogg supersingular) - $m\_\tau/m\_\mu \approx \text{Ogg17} - \pi/(N\_c{\cdot}C\_2)$ (Lyra; substrate-natural form with forward prediction; 0.008% match) - 3 independent Ogg17 cross-context appearances Wednesday (Lyra m_τ/m_μ + Grace 97/98 + Elie Macdonald $-136/135 = -(\text{rank}^3{\cdot}\text{Ogg17})/(N\_c^3{\cdot}n\_C)$) - Monster Moonshine connection (Ogg supersingular primes are the supersingular j-invariant primes; substrate’s lepton observables naturally use Moonshine prime structure) - Pin(2) cover on S^1 provides spinor substrate content where leptons live

Falsification gate: any lepton-sector observable that operates exclusively via Mersenne structure (not Ogg/Moonshine) would refine OMAP-Shilov scope.

5. Paired sister-principles structure

TMAP-Bulk and OMAP-Shilov are not competing — they are paired: - They operate in DIFFERENT regions (substrate-spatial complementarity) - They use DIFFERENT prime sets (substrate-arithmetic complementarity)

- They predict DIFFERENT observables (substrate-physics complementarity) - Together they characterize substrate's full operational mathematics

Common backbone: both principles operate within substrate Hall algebra (Casey directive #4) at q=2 Macdonald specialization (Lyra v0.6). The two regions correspond to DIFFERENT specializations of t parameter: - Bulk: q=2, t=? (Mersenne emphasis; possibly t=1 Hall-Littlewood limit) - Shilov: q=2, t=α=1/N_max (Ogg/Moonshine emphasis; α-deformation per N_max=137 Dirac critical)

This is hypothesis — t-parameter specialization per region needs explicit Hall algebra derivation (Phase 0 closure work).

6. Concrete observable assignments

6.1 OMAP-Shilov region

Observable	Substrate form	Tier
m_τ/m_e	$g^2 \cdot$ Ogg71 = $49 \cdot 71 = 3479$	T2003 RATIFIED
m_μ/m_e	$(24/\pi^2)^6 \approx 207$ (or refined per Grace (rank^2/ζ(2))^c_2)	T190 RATIFIED
m_τ/m_μ	$\approx$ Ogg17 $- \pi$/(N_c·C_2) (forward prediction)	FRAMEWORK-PLUS Wednesday
Electron K-type	V_(1/2, 1/2) Bergman ρ-weight (N_c, rank)	RATIFIED Memorial Day
Photon K-type	V_(1,0) Wallach dim_5 (catalog)	RATIFIED
a_e (anomalous magnetic moment)	Schwinger $\alpha/(2\pi)$ + cover-required structure	T2476 RATIFIED
Five-Absence (Shilov-side)	NO sterile neutrino (Ogg structural absence)	K65 RATIFIED

6.2 TMAP-Bulk region

Observable	Substrate form	Tier
m_b/m_d	g · M_g = $7 \cdot 127 = 889$	FRAMEWORK-PLUS Wednesday
m_p/m_e	$6\pi^5 = 1836.12$ (0.002%)	T187 RATIFIED
Quark mass hierarchy	Cyclotomic chain at q=2 Mersenne structure	Phase 0 dependent
Nucleon (3 quarks)	Bulk K-type configuration in color singlet	Memorial Day vision
QCD β_0	(11N_c $-$ 2N_f)/(12π) — substrate gauge structure	T-RATIFIED
Λ_QCD	Inverse T^2 circumference in bulk confinement	SP-26 W-18
Five-Absence (Bulk-side)	NO proton decay (chain termination structural absence)	K65 RATIFIED

7. Cross-CI work implications

7.1 Lyra theoretical

- Investigate which Hall algebra subalgebra corresponds to TMAP-Bulk vs OMAP-Shilov regions
- Derive Macdonald specialization (q=2, t-bulk) vs (q=2, t-Shilov=α) per region
- Forward-derive specific lepton + quark observables from regional substrate-mechanisms
- Substrate Hall Algebra paper #388: present TMAP-Bulk + OMAP-Shilov as paired sister principles

7.2 Elie compute

- Verification toys: Mersenne predictions in bulk-region observables (TMAP-Bulk tests)
- Verification toys: Ogg/Moonshine predictions in Shilov-region observables (OMAP-Shilov tests)
- Cross-region verification: predict NEW observable in each region; check against catalog or measurement

7.3 Grace catalog

- Reorganize catalog by REGION TAG (Shilov / Bulk / boundary-spanning)
- Enumerate Mersenne substrate-natural observables (TMAP-Bulk catalog)
- Enumerate Ogg/Moonshine substrate-natural observables (OMAP-Shilov catalog)
- Cross-reference 228 Mersenne entries (Wednesday Grace) → TMAP-Bulk
- Cross-reference Ogg supersingular entries → OMAP-Shilov

7.4 Cal verification

- Type C tier check (Cal #122) on bulk-vs-Shilov substrate-mechanism distinction itself
- Tier discipline on TMAP-Bulk + OMAP-Shilov paired-principle filing
- Cal #29 question-shape audit: forward enumeration of regional assignments (not back-fit from observed prime appearances)
- Cal #133 partial-tautology check: small-prime coincidence vs substantive substrate-mechanism

7.5 Keeper synthesis

- This document (v0.1) provides the formal framing
- Update Investigation Board v0.2 with bulk-vs-Shilov as primary structural framework
- Coordinate Substrate Hall Algebra paper structure with Lyra (paired sister principles as content)
- Cal #30 + Cal #31 candidate filing coordination if triggered

8. Open verification gates

For TMAP-Bulk Casey-named promotion: 1. Universal-in-bulk verification: enumerate quark + nucleon + heavy-composite observables; verify Mersenne substrate-mechanism applies to ≥80% (Cal #29 threshold analog) 2. Substrate-mechanism for Mersenne emphasis in bulk: WHY does bulk use Mersenne specifically? Candidate: K59 RS coding on GF(2^g) is bulk substrate-computational backbone 3. No-Ogg-in-bulk verification: bulk observables should NOT operate via Ogg/Moonshine structure (otherwise distinction weakens)

For OMAP-Shilov Casey-named promotion: 1. Universal-in-Shilov verification: enumerate lepton + light-boson observables; verify Ogg/Moonshine substrate-mechanism applies to ≥80% 2. Substrate-mechanism for Ogg emphasis on Shilov: WHY does Shilov use Ogg specifically? Candidate: Monster Moonshine + Pin(2) cover content on S^1 3. No-Mersenne-in-Shilov verification: Shilov observables should NOT exclusively operate via Mersenne

Paired gate: cross-region predictions distinguishable — each principle predicts where the OTHER doesn't apply. Falsifiability via region-specific predictions.

9. Tier disposition

- Bulk-vs-Shilov substrate-mechanism distinction: FRAMEWORK-PLUS per Casey directive 2026-05-27 PM elevation
- TMAP-Bulk principle: Casey-named CANDIDATE pending substantive verification (Gate 1-3 above)
- OMAP-Shilov principle: Casey-named CANDIDATE pending substantive verification (Gate 1-3 above)
- Paired-principles formal structure: v0.1 position document filed for cross-CI coordination
- Substrate Hall algebra at differentiated specialization per region: HYPOTHESIS at INTERPRETIVE tier; Phase 0 closure derivation pending

No premature RATIFIED. Cal #27 STANDING + Cal #29 STANDING + Cal #133 partial-tautology distinction applied throughout. Cal cold-read on this paired-principle filing queued.

10. Connection to existing structure

Connected element	Status
Bulk-Boundary Two-Face Structure of Integer-Webs (Task #237, Casey vision pending)	ELEVATED to in_progress Wednesday PM — this document operationalizes
Three lepton generations refined candidate (Task #384)	OMAP-Shilov subset; chain levels X=N_c, n_C, g in Shilov region
Quark sector substrate-natural investigation (Task #386 new)	TMAP-Bulk substantive investigation
Multi-phase quiver Hall algebra at q=2 (Lyra v0.6)	Common backbone; t-specialization differs per region
Substrate Hall Algebra paper (Task #388 primary)	Includes TMAP-Bulk + OMAP-Shilov as content
Manifolds-require-boundaries framework (Task #371)	Bulk-vs-Shilov is the operational manifestation
Casey Memorial Day “S^1 enacts physics” intuition	Refined: S^1 enacts Shilov-region physics via OMAP; bulk enacts via TMAP
Cal #139 cyclotomic chain	TMAP-Bulk backbone (q=2 q-integer Mersenne)
Grace 9-element substrate operational arithmetic set INV-5195	UNION of TMAP-Bulk (Mersenne) + OMAP-Shilov (Ogg) ingredient sets

11. Standing for next moves

This document provides v0.1 framing for Lyra’s investigation track (Tasks #385 bulk-vs-Shilov, #386 quark sector, #387 paired sister principles, #388 substrate Hall algebra paper) + Grace catalog reorganization by region + Elie regional verification toys + Cal tier discipline review.

Casey decision queued when paired principles substantively pan out: - Promote TMAP-Bulk to Casey-named principle? - Promote OMAP-Shilov to Casey-named principle? - Promote bulk-vs-Shilov distinction itself to Casey-named principle? - Or all three? (Could be Casey-named principles #9, #10, #11 if all promote)

— Keeper, 2026-05-27 Wednesday afternoon